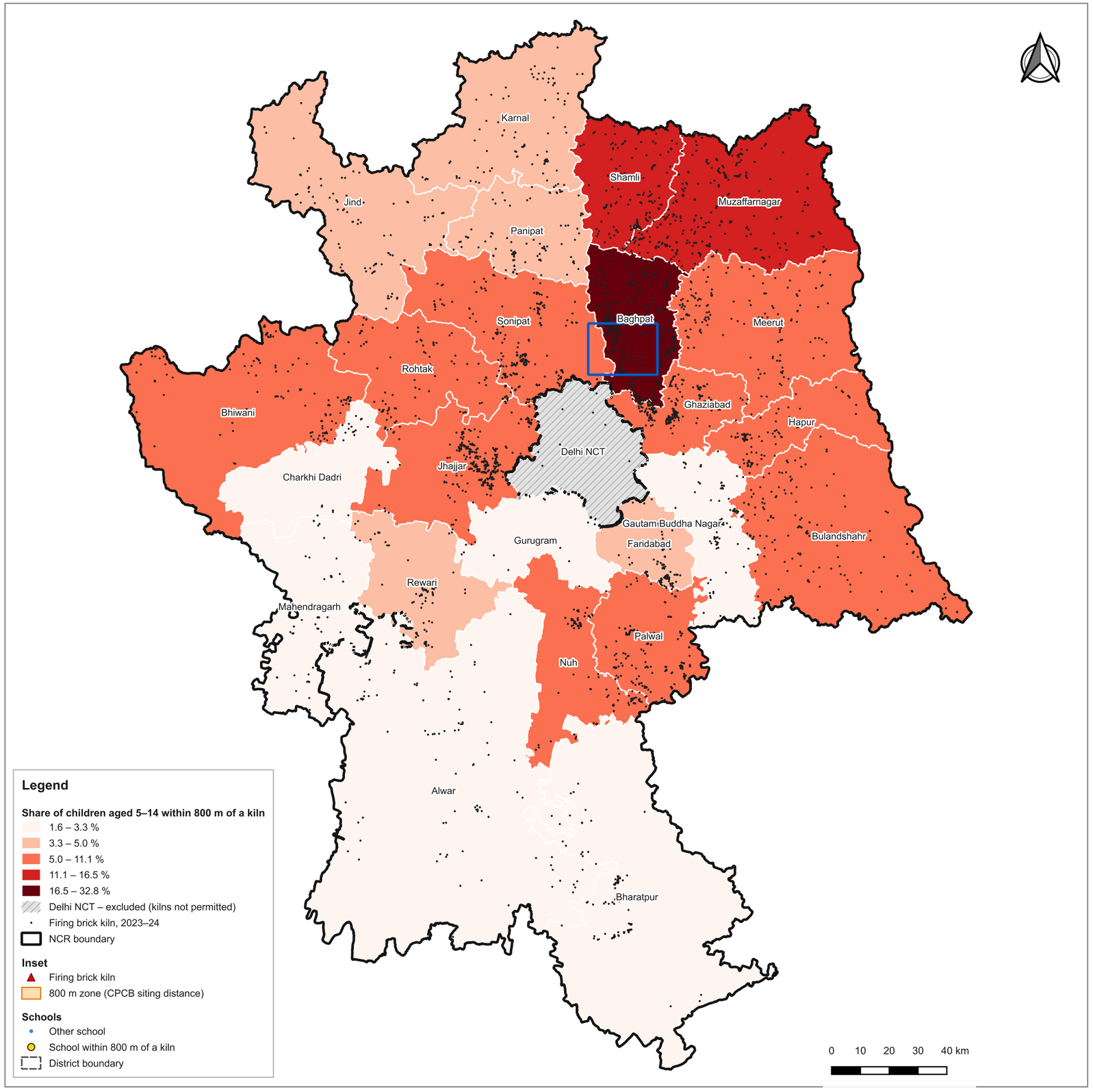
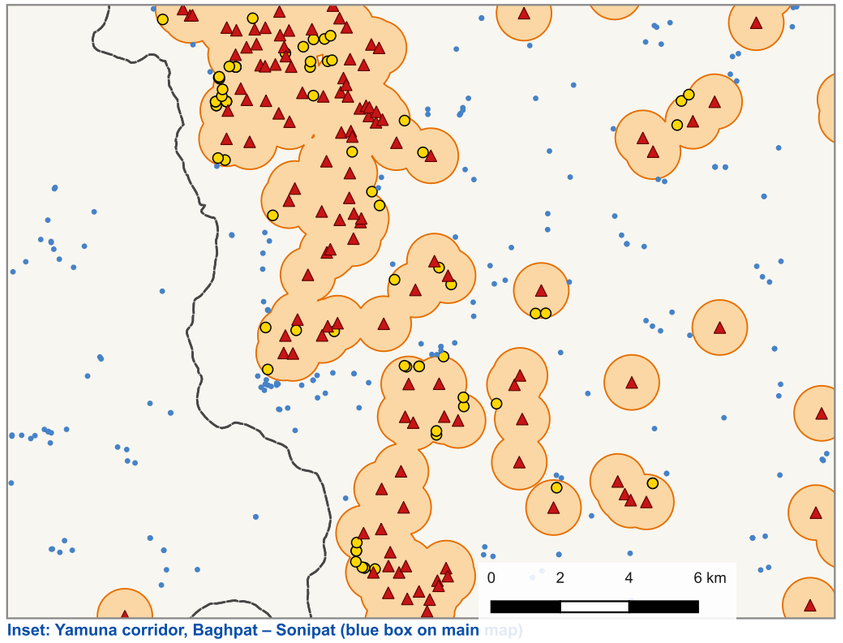
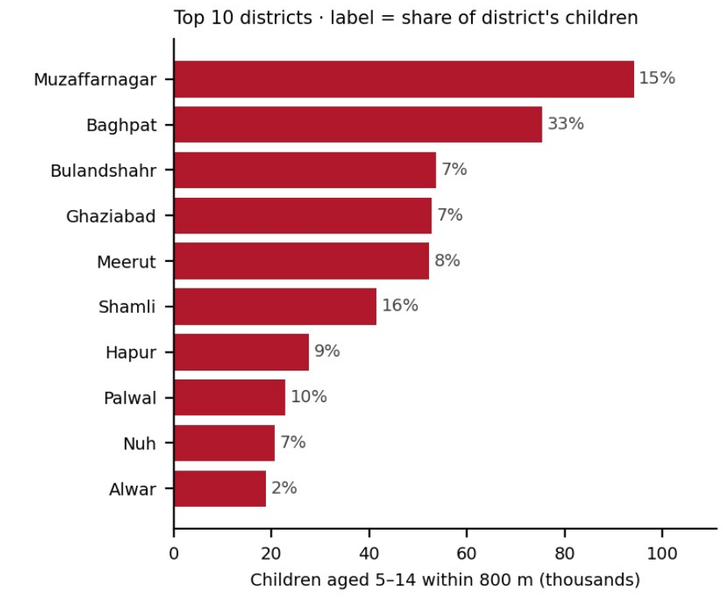
One in three children in Baghpat lives within 800 m of a firing brick kiln

Brick kilns found by AI on Sentinel-2 imagery, and the children and schools inside the CPCB 800 m siting distance · National Capital Region, 2023–24 firing season





Inset: Yamuna corridor, Baghpat – Sonipat (blue box on main map)



Key numbers (NCR excluding Delhi)

≈ 605,000 children aged 5–14 (7.3%) live within 800 m of a firing kiln
1,708 schools stand within 800 m · 35% of 3,256 kilns have a school within 800 m
Kilns near schools were no more likely to stop firing by 2025–26 (+1.3 pts, p = 0.31)

Method: YOLO11 oriented-box detector on Sentinel-2 winter composites, filtered by land cover, verified by SWIR hot-spot firing frequency, non-kiln heat sources removed. Precision 0.91 by visual audit of 25 random kilns (95% CI 0.73–0.98). Children: WorldPop 2020, ages 5–14. Schools: UDISE (DataMeet), location-checked.
Data: Copernicus Sentinel-2 (ESA), SentinelKilnDB (IIT Gandhinagar, CC BY-NC 4.0), ESA WorldCover, WorldPop, UDISE/DataMeet, OpenStreetMap contributors, NCRPB boundaries (rebuilt).
CRS: WGS 84 / UTM zone 43N (EPSG:32643) · Produced 13 October 2026 · KilnWatch – Rashmi Sinha